

# Which Hand Gestures Are Natural?

A sensor-agnostic map of one hand’s gesture space, alone and in combination

Ethan Lambeth

Working paper · <https://ethanlambeth.com/natural-hand/>

Model v2.0.0 · results generated 2026-10-07

## Abstract

Gesture vocabularies for glasses, wristbands and rings are usually chosen from what a sensor can detect, then checked for comfort. We invert the order and ask what the hand itself finds natural, before any sensor or meaning is considered. We generate the hand’s gesture space from its own degrees of freedom—148 gestures across thumb contacts, single-digit movements, whole-hand synergies, handshapes, wrist and forearm axes and isometric force—and score each resolved joint trajectory with seven costs: five from laboratory evidence (posture, coupling, thumb reach, endurance, effort) and two from a new analysis of everyday hand use (1.54 h of 22 people performing 26 daily activities). The everyday data replicate the published synergies of natural hand movement (cosines 0.97 and 0.97), and everyday rarity alone tracks rated thumb-contact discomfort ( $\rho = 0.73$ ,  $n = 15$ ), ratings the everyday terms never see (the thumb-reach term is built from them, so this tests the everyday half only). The ranking is stable under joint uncertainty in weights, assumptions, resting posture and participants (median  $\rho = 0.96$ ). Small wrist and forearm movements, partial opening of the hand and lone thumb movements are the most natural; among contacts, the thumb on the distal and middle segments of the index leads, and the ring and little fingers and the marked handshapes come last. Fluent sequences change cost little: repeating a gesture flows, moving between ring and little contacts clashes. Doing two gestures at once always costs more than the harder one alone. Of six registered predictions from independent studies, the model passes three and fails three. One failure (large movements of the ring and little fingers) prompted a fix from its own source; it still fails, now because bounded cost terms saturate, and it no longer counts as an independent test.

## 1 Introduction

Smart glasses, EMG wristbands and rings all accept hand gestures. Their vocabularies converge on a few conventions—an index pinch to select, a held pinch to drag, a wrist turn to adjust—because those are what current sensors detect reliably. That order of reasoning, sensor first and body second, makes it hard to tell which conventions are natural for the hand and which are artefacts of the sensor.

This paper isolates the first question: *which hand gestures are most natural for the human hand, on their own and in combination?* Sensing (what a camera, band or ring can see) and meaning (what people expect a gesture to do) are deliberately excluded. They are filters to apply afterwards to a canonical, body-only ranking, not inputs to it.

### Contributions.

1. **A generative gesture space** for one hand: every thumb-to-finger contact (finger  $\times$  segment  $\times$  surface  $\times$  action), single-digit curls and lifts, whole-hand synergies, 14 handshapes, six wrist and forearm directions and isometric force, with free movements at three amplitudes (148 gestures; 81 reachable at the canonical amplitude).

2. **A cost model in two independent parts:** laboratory evidence on comfort and control, and everyday statistics computed here from raw daily-activity kinematics [11]. Contacts are solved by inverse kinematics with an everyday-posture prior, so the thumb’s real joint angles are scored.
3. **Five computational experiments:** replication of everyday synergies; construct validity of everyday rarity against comfort ratings; a robust ranking with uncertainty-based tiers; registered predictions from held-out studies; and scoring of 6,561 sequences and 271 chords as trajectories.
4. **A canonical set** of natural gestures in four robustness tiers, with code, data and an interactive demonstration.

Throughout we separate *reported findings* (numbers from the literature), *assumptions* (coefficients we chose; Appendix C) and *model outputs*. No output here is a measurement of people performing these gestures; Section 6 says what would make it one.

## 2 Evidence

The six-strand review behind this work (approximately 190 sources) is summarised here only where it bears on the body. Its sensing and meaning strands are set aside for later work.

**Independence.** The thumb and index move almost independently; the ring does not. Individuation indices are 0.983, 0.982, 0.937, 0.907 and 0.943 from thumb to little finger [8]. In force, pressing with one finger recruits its neighbours, up to 23% of the receiving finger’s maximum for middle→ring [26]. Enslaving is stronger in extension than in flexion [28]: extending a finger recruits over half of a neighbour’s extensor units [24].

**Posture, effort and endurance.** Discomfort rises in steps at joint-angle class edges [14]. Sign-language interpreters rate slight finger flexion least uncomfortable (0.4 of 5) and claw (2.7), extended (2.5) and spread (1.9) shapes worse; wrist flexion is the strongest predictor of discomfort [19]. Grip endurance follows  $ET = 33.55 f^{-1.61}$  s at a fraction  $f$  of maximum effort [7]; light pinches load the hand at about 4–5% of maximum [21]. Tolerated repetition falls from finger to wrist to elbow in the ratio 6:3:2 [15].

**Thumb-to-finger comfort.** Thumb taps are rated most comfortable on the distal index (4.7 of 5) and middle (4.5) segments and least on proximal segments and the little finger (about 2); slides rate below taps on every segment [9]. Side contacts rate below the adjacent pad [12].

**Natural hand use.** In 17 hours of daily life, two velocity components explain 60.2% of finger movement: a whole-hand open/close and a knuckle-versus-finger-joint bend [10]. Everyday co-use, more than muscle anatomy, explains how the cortex represents the fingers [5], and grasp postures share the same low-dimensional synergies [22]. That motivates scoring gestures against what the hand does every day.

## 3 Method

### 3.1 The gesture space

Gestures are generated from the hand’s degrees of freedom, not from any vocabulary in use (Table 1). All start and end at HOME, the everyday mean finger posture [10] with a neutral wrist and forearm [25], so gestures differ only in what they ask of the hand. Free movements come in

small, medium and large amplitudes (30%, 60% and 100% of the way to their end shape; wrist ranges from the functional range [20]). Cross-family comparisons use the medium amplitude; Section 4.4 tests that choice. Arm travel and two-handed gestures are out of scope: their cost is dominated by the shoulder and by where the arm is, which belongs to the situation, not the hand.

**Table 1:** The generated gesture space.

| Family          | Construction                                                                                                                                                                                                                     | <i>n</i> |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Thumb contacts  | Tip-to-tip pinch (tap, hold) on each finger; thumb pad on the palmar pad of each finger’s three segments (tap, hold, slide along); thumb pad on the index’s radial side at three segments (tap, hold, slide along, slide across) | 56       |
| Single digits   | Each finger curls or lifts alone; the thumb swings across the palm, swings out, or curls; three amplitudes                                                                                                                       | 33       |
| Whole hand      | Close, open, knuckle bend, hook, spread and three finger groups at three amplitudes; adduction                                                                                                                                   | 25       |
| Handshapes      | Unmarked B, A, S, C, O, 1, 5 and marked V, W, Y, I, L, 3, horns [3], held 0.8 s                                                                                                                                                  | 14       |
| Wrist & forearm | Flexion, extension, radial, ulnar, pronation, supination; three amplitudes                                                                                                                                                       | 18       |
| Force           | Fist squeeze; pinch press                                                                                                                                                                                                        | 2        |

### 3.2 Trajectories and contacts

Each gesture is sampled at 80 intervals on a rigged hand (25 joints). Thumb contacts are solved by inverse kinematics: the thumb pad must land on the target, and among the ways to do so the solver prefers the posture closest to everyday use (squared Mahalanobis distance under the everyday covariance, traded at 16 units per millimetre of miss). The touched finger may curl to meet the thumb. A contact counts as reachable if the solved pad lands within 5 mm. 17 contacts miss by more, all but one on proximal segments; they are reported but not ranked (Appendix B).

Each frame is mapped to the 17 anatomical angles of the everyday dataset (thumb CMC abduction and flexion, thumb MCP and IP, finger MCP and PIP, three relative abductions, wrist flexion and deviation). Finger and wrist angles map directly. The thumb’s CMC angles are measured geometrically from the rig’s first metacarpal against the fixed second metacarpal; because the rig and the glove do not share a thumb-CMC zero, both are anchored by matching HOME to the everyday mean. Forearm rotation and the finger DIP joints are not in the everyday data and are scored by the laboratory terms only.

### 3.3 Everyday statistics

The KIN-MUS UJI dataset records 22 right-handed adults performing 26 activities of daily living with a calibrated 18-angle glove at 100 Hz [11]. We drop the palmar arch (the rig cannot express it), remove frames with implausible wrist angles (9% of frames; the dataset authors flag spurious extension), smooth with a 50 ms window and differentiate. Statistics weight subjects equally: the pose mean and covariance combine within- and between-subject variance, and the velocity covariance averages subjects’ second moments. This yields 502,727 frames (1.54 h).

### 3.4 Cost terms

All terms lie in  $[0, 1]$ , where 0 is free. Means are taken over frames where the hand has left HOME, so idle time does not dilute a gesture.

**Laboratory terms.**

- **Posture** [14, 19]: per frame,  $0.65 S + 0.85 W$ .  $S$  places the finger shape on the interpreters' 0–5 scale (slight flexion to claw), adds the thumb's distance from neutral when it is not in contact, and divides by 5.  $W$  sums, per wrist and forearm axis, the larger of the interpreters' rating above neutral and the LUBA class score scaled to 0.6. The term is 70% peak and 30% mean.
- **Coupling** [8, 16, 26, 28]:  $0.55 (L/F)/0.10 + 0.45 D$ , where  $L/F$  is joint travel weighted by  $1 - \Pi$  of the moving digit as a share of all travel, and  $D$  is the enslaving-weighted discordance between finger flexions beyond that at HOME. Extension of the middle, ring or little finger counts  $1.5\times$ , and each digit's leak grows with its distance from HOME,  $\times(1 + a)$  where  $a$  is that distance as a fraction of everyday range (added after the first outputs; Section 6).
- **Thumb reach** [9, 12]: contact-weighted mean of  $(5 - c)/4$ , with  $c$  the tap or stroke comfort of the touched segment, plus 0.08 on the side surface.
- **Endurance** [7, 21]:  $1 - \exp(-\sum 60 h/ET(f))$  over holds of  $h$  seconds, at  $f = 0.02$  (free), 0.04 (contact) or 0.25 (force).
- **Effort** [15]: each digit, the wrist and the forearm count their travel as a fraction of everyday range (5th–95th percentile; DIP and forearm from functional ranges [2, 18]), averaged over the segment's joints and weighted 1:2:3 for finger, wrist and forearm; cost  $1 - \exp(-x/4)$ .

#### Everyday terms.

- **Everyday pose**: the share of everyday time the hand is closer to its mean shape than this frame (empirical distribution of the Mahalanobis distance); 70% peak, 30% mean. HOME itself scores 0.09.
- **Everyday movement**: for each frame-to-frame velocity direction  $\hat{v}$ ,

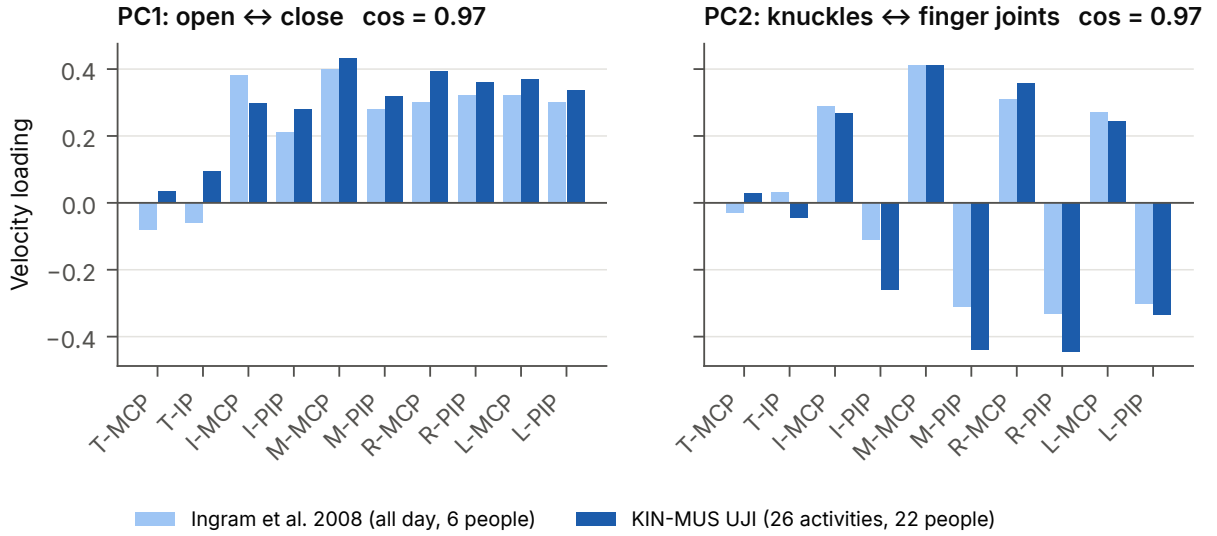
$$m(\hat{v}) = \frac{\log(\hat{v}^\top \tilde{\Sigma}^{-1} \hat{v} \lambda_{\max})}{\log(\lambda_{\max}/\lambda_{\min})},$$

where  $\tilde{\Sigma}$  is the everyday velocity covariance plus a ridge of 1% of its largest eigenvalue: 0 for the most common direction of movement, 1 for the rarest. The term is the energy-weighted mean. Pure forearm rotation leaves it undefined.

The composite is the equally weighted mean of the defined terms. Weights are not fitted, because no study rates these gestures on one scale. Equal weights privilege no term; Section 4.7 shows the ranking barely depends on them.

### 3.5 Combinations

Combinations are scored as trajectories by the same model. *Sequences*  $A \rightarrow B$  are performed fluently: B's core begins while the last quarter of A's is finishing, and the ready posture glides from A's to B's. Their *flow interaction* is the cost of that trajectory minus the cost of the same two gestures separated by a return to HOME. *Chords*  $A + B$  hold A (a contact hold or handshape) while B is performed on top of it, by superposing joint displacements. A chord is impossible when both gestures need the thumb, both move the same digit, or one moves the finger the thumb is touching. Its *chord penalty* is the cost of the chord minus the cost of the harder gesture alone.



**Figure 1:** First two everyday velocity components: published all-day loadings [10] against the activities-of-daily-living data analysed here [11].

### 3.6 Registered predictions

Before the model produced any output, we registered twelve predictions in code (Table 3). Six are *independent*: their source shares no data with any term. Four are *convergent*: they restate a construct an input already encodes, so passing them shows consistency, not validity. Two conflict with each other: comfort ratings prefer the tip pinch [13], while everyday grasping uses the lateral pinch 30 times more often [6]. A thirteenth prediction (thumb abduction fatigues more slowly than flexion) was dropped before computing because the rig cannot express palmar abduction. Items the rig cannot reach are untestable and skipped.

## 4 Results

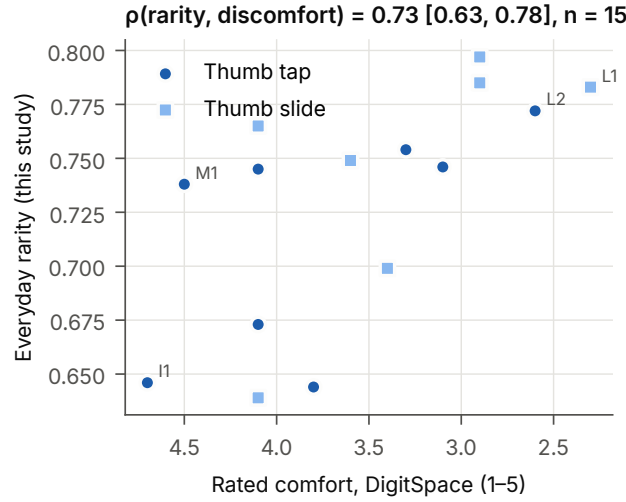
All numbers below are model outputs.

### 4.1 The everyday data replicate natural-hand synergies

On the ten joints the two datasets share, the first two velocity components of everyday activity match Ingram et al.’s all-day components with cosines of 0.97 and 0.97 (Figure 1), and the planes they span overlap by 0.96 (95th percentile for random planes: 0.40). They explain 65% of velocity variance on those joints, against 60.2% for Ingram’s 19 joints. A single digit moves alone in 14% of moving time and all four fingers move together in 50%—more coupled than all-day life (7% and 17%; [10]), as goal-directed manipulation would predict. The two sources describe the same structure, and the everyday terms rest on it.

### 4.2 Everyday rarity predicts rated comfort

The everyday terms never see a comfort rating, so they can be tested against them. We report every validity test as the rank correlation between everyday rarity and rated *discomfort*, so a positive  $\rho$  always means agreement. Across the 15 reachable thumb tap and slide conditions rated in DigitSpace,  $\rho = 0.73$  (95% subject-bootstrap interval 0.63 to 0.78; permutation  $p = 0.002$ ; Figure 2). This tests the everyday terms only: the composite also contains the thumb-reach term, which is built from these ratings. Thumb contacts the hand rarely makes in daily life are the ones people rate uncomfortable. The test is weaker elsewhere. Across the interpreters’ five



**Figure 2:** Everyday rarity (this study) against rated thumb-contact comfort (DigitSpace Figure 2 means).  $\rho$  is reported between rarity and discomfort, so positive means agreement. Labels: finger and segment (1 distal, 2 middle).

finger shapes  $\rho = -0.10$ , a failure: an extended hand is common in daily activity but rated uncomfortable. Across four wrist postures  $\rho = 0.63$ , in the expected direction, but with four items its bootstrap interval is uninformative.

### 4.3 The canonical ranking

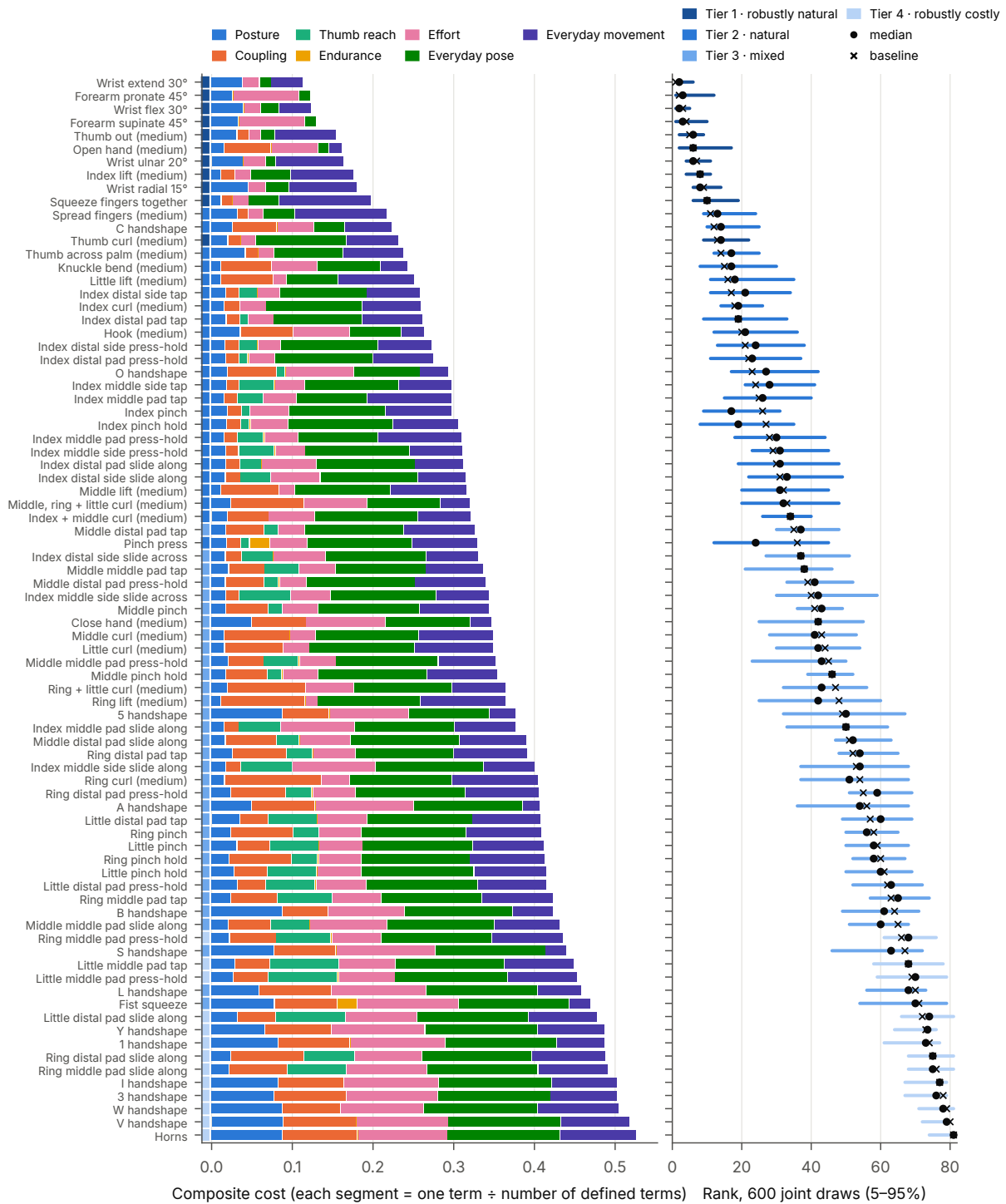
Figure 3 decomposes the cost of the 81 reachable gestures at the canonical amplitude, alongside their ranks under joint uncertainty (Section 4.7). Tiers follow from that uncertainty, not from cost thresholds: Tier 1 gestures rank in the top quarter in at least 90% of draws, Tier 2 in the top half in at least 75%, Tier 4 in the bottom quarter in at least 90%; Tier 3 is the rest (Table 2).

Four patterns stand out.

1. **Proximal and gentle movements lead.** Wrist and forearm movements of 15–45° and partial opening of the hand cost least. They involve no coupling, no contact and few joints, and the everyday data show them to be common directions of movement. Their main cost is posture, which the interpreters' ratings put at 0.27–0.32 for these wrist movements.
2. **The thumb is the freest digit.** Moving the thumb out or curling it are Tier 1 and swinging it across the palm is Tier 2, consistent with its near-perfect individuation [8].
3. **Among contacts, the index's distal and middle segments lead.** The best contacts are the thumb pad on the distal side or pad of the index (Figure 4); the tip-to-tip index pinch (0.299) and the lateral (key) pinch on the side of its middle segment (0.298) are tied. Cost rises steadily from index to little finger and from distal to middle segment, and slides cost more than taps and holds. All contacts carry high everyday-pose costs (0.61–0.98): pressing the thumb against a finger is uncommon in daily activity, which is mostly grasping.
4. **Ring, little and marked shapes trail.** Contacts on the ring and little fingers and the marked handshapes (V, W, I, Y, 3, horns, 1) fill Tier 4 and the bottom of Tier 3; the ring finger moved alone sits low in Tier 3.

### 4.4 Amplitude

Cost rises with amplitude for every free movement (Figure 5), so the most natural version of any movement is the smallest that still counts as deliberate. The order of gesture types barely



**Figure 3: Canonical ranking.** Left: composite cost, decomposed into the seven terms (each term's share is its value divided by the number of defined terms); the strip marks the tier. Right: rank across 600 joint draws of weights, assumptions, resting posture, everyday-data bootstrap and IK solution. Gesture names in Table 2 and Appendix A link to the demo.

**Table 2:** The canonical set: 81 gestures in four robustness tiers, in rank order within each tier and family (model output).

| Tier | <i>n</i> | Gestures                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 11       | <i>Wrist &amp; forearm:</i> Wrist extend 30°, Forearm pronate 45°, Wrist flex 30°, Forearm supinate 45°, Wrist ulnar 20°, Wrist radial 15°; <i>Single digits:</i> Thumb out (medium), Index lift (medium), Thumb curl (medium); <i>Whole hand:</i> Open hand (medium), Squeeze fingers together                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2    | 24       | <i>Whole hand:</i> Spread fingers (medium), Knuckle bend (medium), Hook (medium), Middle, ring + little curl (medium), Index + middle curl (medium); <i>Handshapes:</i> C handshape, O handshape; <i>Single digits:</i> Thumb across palm (medium), Little lift (medium), Index curl (medium), Middle lift (medium); <i>Thumb contacts:</i> Index distal side tap, Index distal pad tap, Index distal side press-hold, Index distal pad press-hold, Index middle side tap, Index middle pad tap, Index pinch, Index pinch hold, Index middle pad press-hold, Index middle side press-hold, Index distal pad slide along, Index distal side slide along; <i>Force:</i> Pinch press                                                                                                                                                                   |
| 3    | 33       | <i>Thumb contacts:</i> Middle distal pad tap, Index distal side slide across, Middle middle pad tap, Middle distal pad press-hold, Index middle side slide across, Middle pinch, Middle middle pad press-hold, Middle pinch hold, Index middle pad slide along, Middle distal pad slide along, Ring distal pad tap, Index middle side slide along, Ring distal pad press-hold, Little distal pad tap, Ring pinch, Little pinch, Ring pinch hold, Little pinch hold, Little distal pad press-hold, Ring middle pad tap, Middle middle pad slide along; <i>Whole hand:</i> Close hand (medium), Ring + little curl (medium); <i>Single digits:</i> Middle curl (medium), Little curl (medium), Ring lift (medium), Ring curl (medium); <i>Handshapes:</i> 5 handshape, A handshape, B handshape, S handshape, L handshape; <i>Force:</i> Fist squeeze |
| 4    | 13       | <i>Thumb contacts:</i> Ring middle pad press-hold, Little middle pad tap, Little middle pad press-hold, Little distal pad slide along, Ring distal pad slide along, Ring middle pad slide along; <i>Handshapes:</i> Y handshape, 1 handshape, I handshape, 3 handshape, W handshape, V handshape, Horns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

depends on the amplitude chosen for comparison: ranks at small and large amplitude correlate with those at medium amplitude at  $\rho = 0.97$  and  $0.97$ . Handshapes are full-range by definition and are compared against medium amplitudes, so their cross-family placement should be read with that in mind.

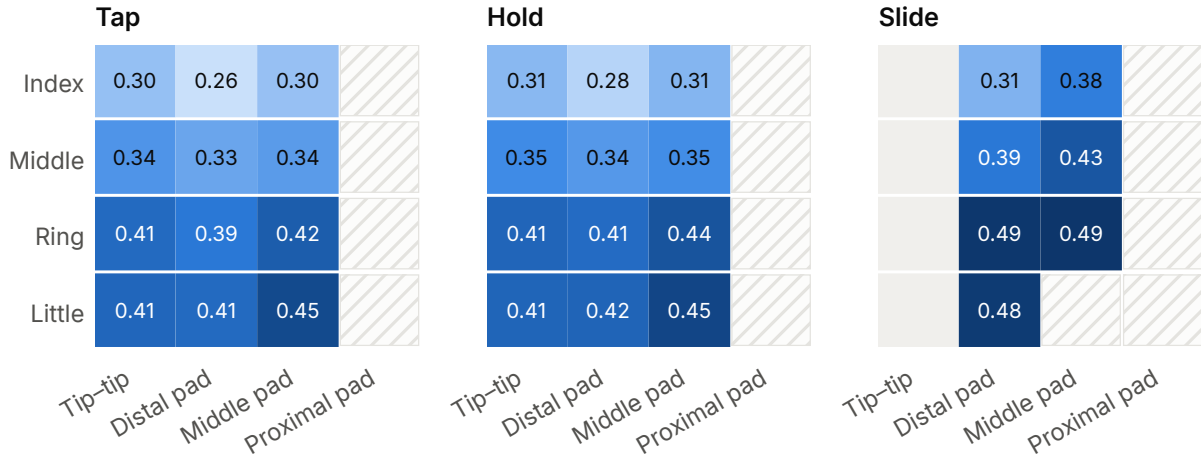
#### 4.5 Laboratory and everyday evidence agree where both apply

The two halves of the model come from unrelated data, so their agreement is informative (Figure 6). Overall they correlate at  $\rho = 0.49$ . They agree strongly on single digits ( $\rho = 0.86$ ), handshapes (0.71) and contacts (0.69). They disagree on whole-hand movements ( $\rho = -0.27$ ) and the wrist (-0.46): closing the hand or hooking the fingers follows the most common everyday directions of movement but costs posture and effort in the laboratory terms. A scale-free consensus—the mean of each gesture’s per-term rank—orders gestures almost exactly as the composite does ( $\rho = 0.94$ ).

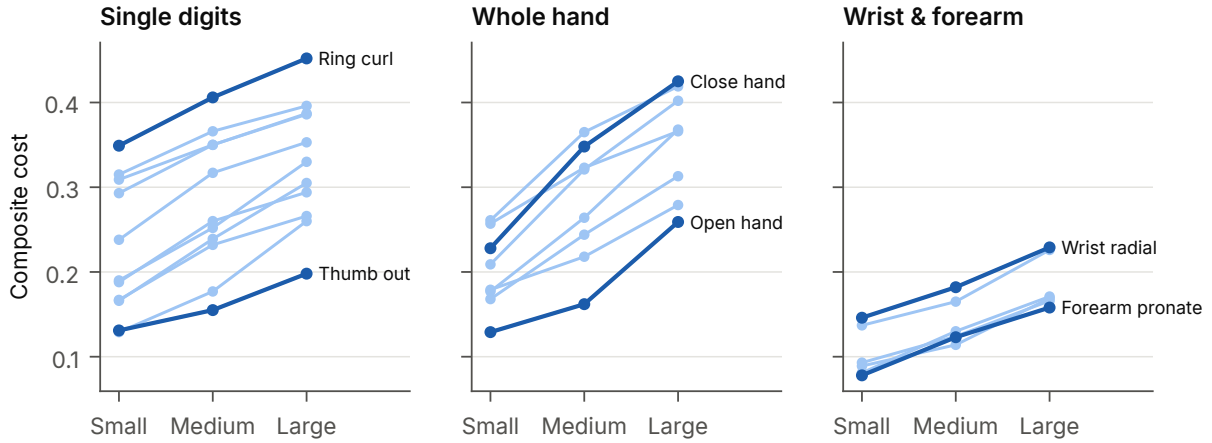
#### 4.6 Registered predictions

Table 3 reports every registered prediction. The model passes 3 of 6 independent predictions (one of which, marked †, has since informed the model) and 4 of 4 convergent ones.

The passes are substantive. The model recovers Aoki et al.’s full tapping order—index, middle, little, ring—from independence, enslaving and everyday data, without seeing tapping speed [1]. It separates unmarked from marked handshapes almost perfectly: a random unmarked shape costs less than a random marked one with probability 0.98 [27]. Ring–little alternation is its worst-flowing pair of fingers, although the six alternation interactions differ by less than 0.004, so both alternation results are weak.



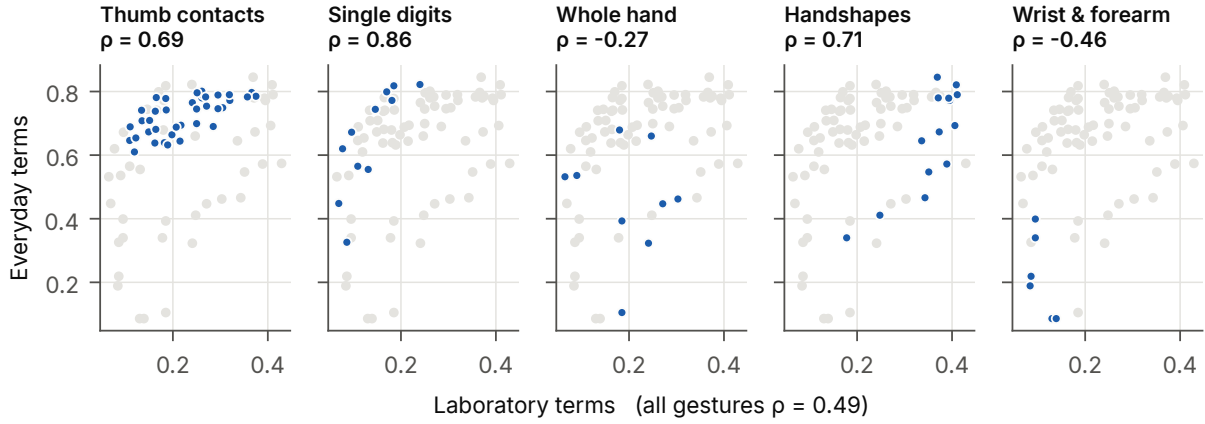
**Figure 4:** Composite cost of thumb contacts by finger, segment and action. Hatched: the rig’s thumb cannot reach within 5 mm. Side contacts on the index appear in Figure 3.



**Figure 5:** Cost against amplitude for each free movement; the cheapest and costliest at large amplitude are labelled.

The failures are informative.

- **Large arcs.** Making an isolated ring or little movement larger should cost disproportionately more than for the index [16]. At first it did not (small-to-large step: index 0.103, ring 0.073, little 0.060), because the coupling term was a *ratio* of leaked to total travel and so did not change with amplitude. We then added the arc factor Lang and Schieber’s result implies († in Table 3), so this prediction no longer tests the model independently. Coupling now grows with the arc (ring curl 0.74 small, 0.98 large), but the prediction still fails (index 0.107, ring 0.103, little 0.076): the everyday-pose term rises steeply for the index, which starts from a typical posture, and is already near its ceiling for the ring and little fingers. Bounded terms that saturate are the remaining limit.
- **Little versus middle.** Moving the little finger alone costs less than moving the middle (5 of 6 pairs hold). The model inherits Häger-Ross and Schieber’s individuation order, in which the little finger (0.943) is more independent than the middle (0.937); SoloFinger’s ease ratings disagree [23].
- **Best alternation.** Index–middle alternation is not the best-flowing pair: non-adjacent pairs (index–ring, index–little) flow better, because their everyday velocities are less correlated. Aoki et al. report index–middle as best; their protocol held the other fingers on



**Figure 6:** Laboratory against everyday composite, by family (other families in grey).

keys, which the model does not represent.

The conflicting pinch pair is a tie (index pinch 0.299, lateral pinch 0.298; tip ahead in 89% of draws): comfort ratings and everyday frequency pull in opposite directions, and a model that uses both kinds of evidence does not resolve them.

#### 4.7 Robustness

The ranking is insensitive to every single assumption (Table 4; the lowest  $\rho$  is 0.95, when the effort term is dropped). The contact solver matters as little: re-solving every contact with a third or three times the everyday-prior weight, or a quarter or four times the finger-curl penalty, leaves  $\rho \geq 0.99$ , and the reach threshold only changes which contacts are ranked. We also sampled the IK solution itself: 20 replicates that pick at random among near-optimal ways to make each contact (objective within  $1 \text{ mm}^2$  of the best, from random multi-starts), with both solver weights drawn over  $\times \frac{1}{3}$ – $\times 3$ , correlate with the baseline at median  $\rho = 0.98$  (lowest 0.97). Some replicates miss the ring and little middle-segment pads by up to 11.1 mm; those contacts sit near the 5 mm reach limit in the baseline too. Under 600 joint draws—Dirichlet weights (concentration 5 per term), a subject-level bootstrap of the everyday data, every assumption varied over its range, one of the IK replicates, and three resting postures (neutral; wrist  $16^\circ$  extended, the measured rest [17]; forearm  $30^\circ$  pronated)—the median rank correlation with the baseline is 0.96 (90% interval 0.91–0.98). Leaving out any one participant changes nothing material (lowest  $\rho = 1.000$ ). The middle of the ranking moves most, the top and bottom least (Figure 3).

#### 4.8 Combinations

**Sequences.** Flow interactions are small next to the costs of the gestures themselves: across all 6,561 ordered pairs, 39% flow better than resetting through HOME. Their structure is nonetheless consistent (Figure 7; Table 5). Repeating a gesture flows (mean interaction -0.009, against 0.001 for switching). Moving between related shapes flows—a held contact into a tap on the same target, the Y shape into horns, V into 3—because the shared posture is kept rather than released and rebuilt. Moving the thumb between ring and little contacts clashes: its path between them passes through more flexed postures and runs against the common everyday directions of movement.

**Chords.** Doing two things at once always costs more than the harder one alone (median penalty 0.033). Holding a contact while moving another finger is cheapest (mean penalty 0.021). Holding a shape or a contact during a wrist or forearm movement costs more (0.036 and

**Table 3:** Registered predictions from held-out studies. Pairs: ordered pairs in the predicted direction. Robust: share of joint uncertainty draws in which the prediction holds. †: the model was changed in response to this prediction after it failed, so it is no longer an independent test (*model output*).

| Prediction                                                                                                          | Result | Pairs | Robust |
|---------------------------------------------------------------------------------------------------------------------|--------|-------|--------|
| <i>Independent</i>                                                                                                  |        |       |        |
| Single-finger tapping is easiest with the index, then middle, then little, then ring. [1]                           | Pass   | 6/6   | 83%    |
| Alternating taps flow best between index and middle (of all six finger pairs). [1]                                  | Fail   | –     | 0%     |
| Alternating taps flow worst between ring and little (of all six finger pairs). [1]                                  | Pass   | –     | 100%   |
| Moving the thumb, index or middle alone is easier than moving the ring or little alone. [23]                        | Fail   | 5/6   | 1%     |
| Unmarked handshapes (B, A, S, C, O, 1, 5) cost less than marked ones (V, W, Y, I, L, 3, horns). [27]                | Pass   | –     | 100%   |
| Making an isolated movement larger costs more for the ring and little fingers than for the index. [16] <sup>†</sup> | Fail   | 0/2   | 2%     |
| <i>Conflicting pair</i>                                                                                             |        |       |        |
| A tip-to-tip index pinch is more comfortable than a lateral (key) pinch. [13]                                       | Fail   | –     | 89%    |
| The lateral pinch is the more natural pinch: everyday grasping uses it 30× more often than the tip pinch. [6]       | Pass   | –     | 11%    |
| <i>Convergent</i>                                                                                                   |        |       |        |
| Thumb contacts on the index and middle are more natural than on the ring, and the ring more than the little. [4]    | Pass   | –     | 57%    |
| On every finger segment, a thumb tap is more natural than a thumb slide. [13]                                       | Pass   | 7/7   | 99%    |
| Thumb–index is the most natural tip pinch. [10]                                                                     | Pass   | –     | 99%    |
| The O and A handshapes are among the more comfortable handshapes (below the median). [19]                           | Pass   | –     | 100%   |

0.040). Holding a contact while curling the other fingers costs most (0.049). The costliest chords combine two individually costly parts (a thumb–index hold while the middle, ring and little fingers curl, or a held contact during supination). Their penalty comes from posture, effort and everyday rarity adding up, not from a specific interference. The most natural chords hold a C shape or a distal index contact while the wrist extends or flexes (Figure 8).

## 5 Discussion

**What is natural for the hand.** The evidence supports a short list. Small wrist and forearm movements, partial opening of the hand, the thumb moving on its own, and the thumb touching the distal or middle segment of the index form the natural core. The middle finger is a reasonable second surface; the ring and little fingers, marked handshapes and stacked holds are not. These conclusions survive every uncertainty we could vary.

**Two kinds of evidence, one picture.** Laboratory ratings and everyday statistics were collected for different purposes by different people, yet they agree on single digits, contacts and handshapes, and the everyday terms track thumb-contact discomfort they never saw. That convergence is the main reason to trust the ranking where it holds. Where they diverge—whole-hand closing and the wrist—the disagreement is itself a finding: the hand moves that way all the time, yet ergonomics still charges for the posture.

**The next filters.** This ranking is deliberately silent on detection and meaning. The intended next steps are to filter it by what each sensor can see, then to assign meanings to what survives.

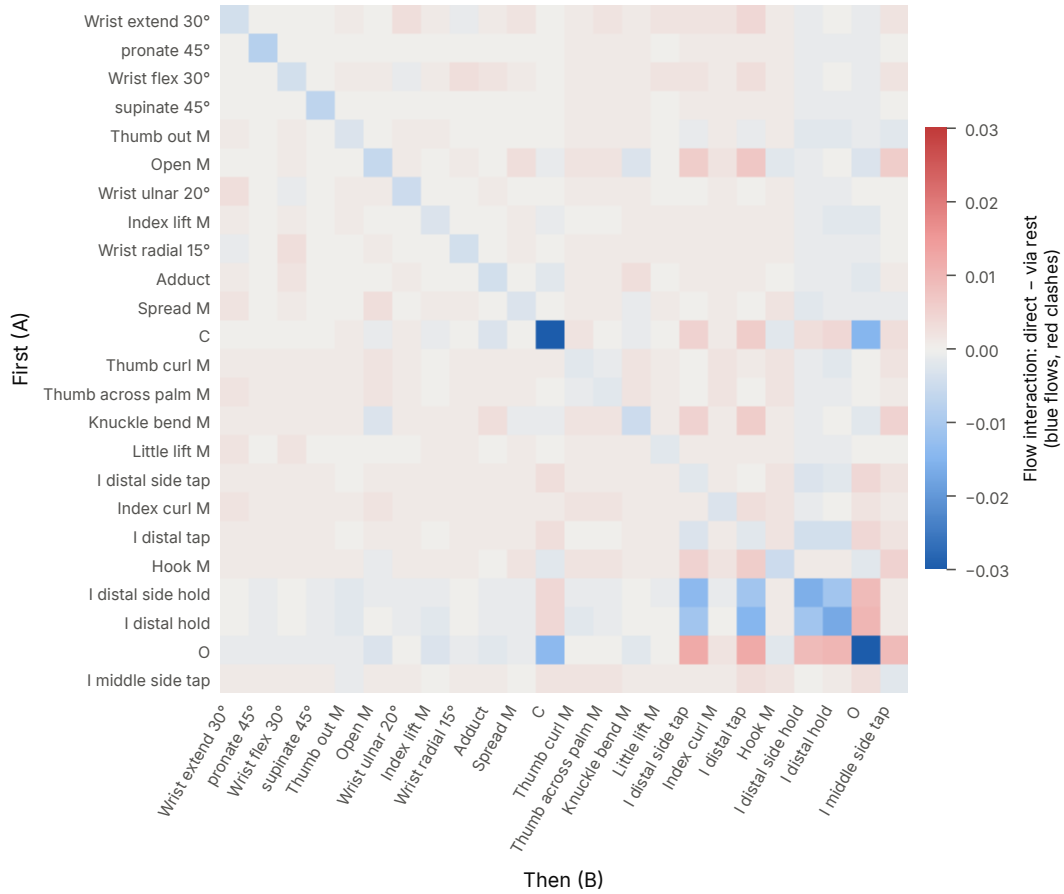
**Table 4:** One-at-a-time sensitivity: Spearman  $\rho$  of the canonical ranking with the baseline (*model output*). Defaults: extension 1.5, side 0.08,  $\varepsilon = 0.01$ , coupling 0.1, arc 1, effort 4, contact 0.04. The last three rows re-solve every contact (prior weight  $10^{-4}$ , curl penalty  $1.6 \times 10^{-5}$ , reach threshold 5 mm).

| Factor                       | Values: $\rho$                                                                                                                       |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Extension leak factor        | 1: 0.998; 1.5: 1.000; 2: 0.997                                                                                                       |
| Side-contact penalty         | 0: 0.999; 0.08: 1.000; 0.16: 0.998                                                                                                   |
| Velocity ridge $\varepsilon$ | 0.001: 0.999; 0.01: 1.000; 0.05: 0.999                                                                                               |
| Coupling scale               | 0.05: 0.989; 0.1: 1.000; 0.2: 0.990                                                                                                  |
| Coupling arc factor          | 0: 0.997; 1: 1.000; 2: 0.997                                                                                                         |
| Effort scale                 | 2: 0.998; 4: 1.000; 8: 0.997                                                                                                         |
| Contact load (MVC)           | 0.02: 1.000; 0.04: 1.000; 0.08: 1.000                                                                                                |
| Drop term                    | Posture: 0.980; Coupling: 0.949; Thumb reach: 0.971; Endurance: 0.999; Effort: 0.962; Everyday pose: 0.977; Everyday movement: 0.973 |
| Resting posture              | extended: 1.000; pronated: 1.000                                                                                                     |
| Wrist artefact filter        | off: 1.000                                                                                                                           |
| IK prior weight              | 0.33x: 1.000; 3.00x: 0.997                                                                                                           |
| Touched-finger curl penalty  | 0.25x: 1.000; 4.00x: 1.000                                                                                                           |
| Reach threshold              | 3 mm: 78 ranked, order unchanged; 8 mm: 82 ranked, order unchanged                                                                   |

The sensing and meaning layers of an earlier version of this model are archived for that purpose.

## 6 Limitations

- **No common-scale validation.** No study has rated these gestures on one scale; every ranking is a model prediction, tested only against partial held-out orderings.
- **Equal weights.** The composite is unfitted. The ranking is insensitive to the weights, but absolute costs are not calibrated to discomfort.
- **Saturating terms.** Coupling now grows with arc size, but bounded terms near their ceiling (everyday pose for the ring and little fingers) still cannot express that large isolated movements of those fingers are disproportionately hard (a failed prediction).
- **Rig and mapping.** Contacts are solved on a generic rig (the ranking is insensitive to the solver’s settings and to near-optimal alternative solutions, but not to the rig itself) whose thumb cannot reach proximal segments and has little palmar abduction; the thumb’s CMC angles are anchored, not calibrated. Everyday-pose costs for contacts depend on this mapping.
- **Everyday data.** KIN-MUS records 1.5 h of goal-directed daily activities, not all-day life, without forearm rotation or DIP joints. Its replication of all-day synergies is strong, but its pose distribution is grasp-heavy.
- **Transferred evidence.** Enslaving forces are applied to motion, interpreters’ ratings to non-signers and occupational repetition data to unloaded gestures. DigitSpace values are the means printed in its Figure 2, to one decimal.
- **Combinations.** Superposition ignores collisions and the extra drive a held digit needs while a neighbour moves, beyond what the coupling term charges.
- **Population.** Sources are mostly young, right-handed adults; strength and age are not modelled.

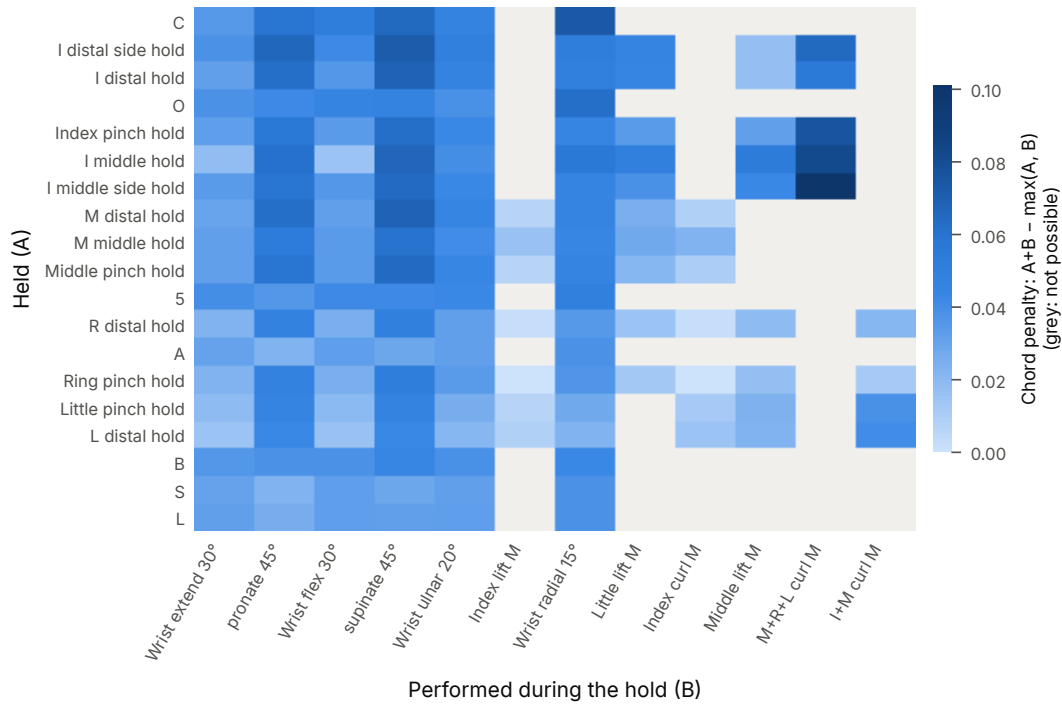


**Figure 7:** Flow interaction for sequences among the leading Tier 1–2 gestures: blue flows better than returning to rest, red clashes.

**Changes after the first outputs.** Four implementation changes were made after the model first ran, and after the registered predictions had been evaluated once; none targeted a prediction. (1) The contact solver’s hand-tuned thumb prior was replaced by the everyday-posture prior, and the touched finger was allowed to curl; this moved the tip/lateral pinch comparison from lateral-ahead to a tie. (2) The effort term had applied Kwon et al.’s per-motion repetition ratios per degree; it now counts range-normalised motions. (3) Sequence overlap had fallen on idle frames of out-and-back movements, so no coarticulation occurred. (4) The reach threshold was raised from 3 to 5 mm after the solver change. (5) After the large-arcs prediction failed, the coupling term gained an arc factor from its source; that prediction is now marked as no longer independent, and it still fails. The registered predictions themselves were not changed; untestable items are skipped. DigitSpace’s values were checked against the numbers printed in its Figure 2; they matched, so nothing changed.

## 7 Conclusion

Ranking gestures by the hand alone, before any sensor, is feasible, reproducible and robust. Laboratory evidence and everyday movement statistics agree on a natural core centred on gentle proximal movements, the free thumb and the distal index. The decisive next step is empirical: one study in which people perform and rate the canonical set, its sequences and its chords on a common scale, which would let the weights be fitted and would test the failed predictions directly.



**Figure 8:** Chord penalty: holding A (rows) while performing B (columns). Grey: impossible (shared digit or thumb).

## References

- [1] Tomoko Aoki, Peter R. Francis, and Hiroshi Kinoshita. Differences in the abilities of individual fingers during the performance of fast, repetitive tapping movements. *Experimental Brain Research*, 152(2):270–280, 2003. doi: 10.1007/s00221-003-1552-z.
- [2] Gregory I. Bain, N. Polites, B. G. Higgs, R. J. Heptinstall, and A. M. McGrath. The functional range of motion of the finger joints. *Journal of Hand Surgery (European)*, 40(4):406–411, 2015. doi: 10.1177/1753193414533754.
- [3] Robbin Battison. *Lexical Borrowing in American Sign Language*. Linstok Press, Silver Spring, MD, 1978.
- [4] Edwin Chan, Teddy Seyed, Wolfgang Stuerzlinger, Xing-Dong Yang, and Frank Maurer. User elicitation on single-hand microgestures. In *Proceedings of CHI '16*, 2016. doi: 10.1145/2858036.2858589.
- [5] Naveed Ejaz, Masashi Hamada, and Jörn Diedrichsen. Hand use predicts the structure of representations in sensorimotor cortex. *Nature Neuroscience*, 18:1034–1040, 2015. doi: 10.1038/nn.4038.
- [6] Thomas Feix, Javier Romero, Heinz-Bodo Schmiedmayer, Aaron M. Dollar, and Danica Kragic. The GRASP taxonomy of human grasp types. *IEEE Transactions on Human-Machine Systems*, 46(1):66–77, 2016. doi: 10.1109/THMS.2015.2470657.
- [7] Laura A. Frey Law and Keith G. Avin. Endurance time is joint-specific: A modelling and meta-analysis investigation. *Ergonomics*, 53(1):109–129, 2010. doi: 10.1080/00140130903389068.
- [8] Charlotte Häger-Ross and Marc H. Schieber. Quantifying the independence of human

**Table 5:** Best-flowing and most clashing sequences, and the most natural and most clashing chords (*model output*). Interaction: sequence minus the same gestures separated by rest; chord minus the harder gesture alone.

| Combination                                                        | Cost  | Interaction |
|--------------------------------------------------------------------|-------|-------------|
| <i>Sequences that flow best</i>                                    |       |             |
| Y handshape → Horns                                                | 0.512 | -0.028      |
| Horns → Y handshape                                                | 0.512 | -0.028      |
| I handshape → Horns                                                | 0.516 | -0.025      |
| Horns → I handshape                                                | 0.516 | -0.025      |
| 3 handshape → V handshape                                          | 0.515 | -0.024      |
| V handshape → 3 handshape                                          | 0.515 | -0.024      |
| Index middle pad press-hold → Index middle pad tap                 | 0.311 | -0.023      |
| Index pinch hold → Pinch press                                     | 0.340 | -0.023      |
| <i>Sequences that clash most</i>                                   |       |             |
| Ring middle pad slide along → Little middle pad tap                | 0.537 | +0.025      |
| Ring middle pad slide along → Little middle pad press-hold         | 0.541 | +0.025      |
| Middle middle pad press-hold → Ring middle pad tap                 | 0.449 | +0.024      |
| Ring distal pad press-hold → Little middle pad tap                 | 0.481 | +0.023      |
| Middle middle pad press-hold → Ring distal pad tap                 | 0.434 | +0.023      |
| Ring middle pad press-hold → Little middle pad tap                 | 0.501 | +0.022      |
| <i>Most natural chords</i>                                         |       |             |
| C handshape + Wrist extend 30°                                     | 0.260 | +0.035      |
| C handshape + Wrist ulnar 20°                                      | 0.271 | +0.047      |
| C handshape + Wrist flex 30°                                       | 0.276 | +0.052      |
| C handshape + Forearm pronate 45°                                  | 0.283 | +0.059      |
| C handshape + Forearm supinate 45°                                 | 0.289 | +0.065      |
| C handshape + Wrist radial 15°                                     | 0.298 | +0.073      |
| Index distal pad press-hold + Wrist extend 30°                     | 0.308 | +0.032      |
| Index distal side press-hold + Wrist extend 30°                    | 0.311 | +0.038      |
| <i>Most clashing chords</i>                                        |       |             |
| Index middle side press-hold + Middle, ring + little curl (medium) | 0.422 | +0.101      |
| Index middle pad press-hold + Middle, ring + little curl (medium)  | 0.403 | +0.082      |
| Index pinch hold + Middle, ring + little curl (medium)             | 0.397 | +0.076      |
| C handshape + Wrist radial 15°                                     | 0.298 | +0.073      |
| Index distal side press-hold + Forearm supinate 45°                | 0.345 | +0.072      |
| Middle middle pad press-hold + Ring + little curl (medium)         | 0.435 | +0.069      |

finger movements: comparisons of digits, hands, and movement frequencies. *Journal of Neuroscience*, 20(22):8542–8550, 2000. doi: 10.1523/JNEUROSCI.20-22-08542.2000.

- [9] Da-Yuan Huang et al. DigitSpace: Designing thumb-to-fingers touch interfaces for one-handed and eyes-free interactions. In *Proceedings of CHI '16*, 2016. doi: 10.1145/2858036.2858483.
- [10] James N. Ingram, Konrad P. Körding, Ian S. Howard, and Daniel M. Wolpert. The statistics of natural hand movements. *Experimental Brain Research*, 188(2):223–236, 2008. doi: 10.1007/s00221-008-1355-3.
- [11] Néstor J. Jarque-Bou, Margarita Vergara, Joaquín L. Sancho-Bru, Verónica Gracia-Ibáñez, and Alba Roda-Sales. A calibrated database of kinematics and EMG of the forearm and hand during activities of daily living. *Scientific Data*, 6:270, 2019. doi: 10.1038/s41597-019-0285-1. Data: doi:10.5281/zenodo.3469380, CC BY 4.0.

- [12] Haiyan Jiang et al. HiFinger: One-handed text entry technique for virtual environments based on touches between fingers. *Sensors*, 19(14):3063, 2019. doi: 10.3390/s19143063.
- [13] Nikhita Joshi et al. Transferable microgestures across hand posture and location constraints. In *Proceedings of UIST '23*, 2023. doi: 10.1145/3586183.3606713.
- [14] Dohyung Kee and Waldemar Karwowski. LUBA: An assessment technique for postural loading on the upper body based on joint motion discomfort and maximum holding time. *Applied Ergonomics*, 32(4):357–366, 2001. doi: 10.1016/S0003-6870(01)00006-0.
- [15] Kwon, You, and Jung. Analysis of maximum acceptable frequencies for upper extremity motions with forces in a seated position. *Journal of the Ergonomics Society of Korea*, 24(1): 65–70, 2005. URL <https://koreascience.or.kr/article/JAK0200516049040623.page>.
- [16] Catherine E. Lang and Marc H. Schieber. Human finger independence: Limitations due to passive mechanical coupling versus active neuromuscular control. *Journal of Neurophysiology*, 92(5):2802–2810, 2004. doi: 10.1152/jn.00480.2004.
- [17] Kyung-Sun Lee and Myung-Chul Jung. Flexion and extension angles of resting fingers and wrist. *International Journal of Occupational Safety and Ergonomics*, 20(1):91–101, 2014. doi: 10.1080/10803548.2014.11077038.
- [18] Bernard F. Morrey, Lawrence J. Askew, and Edmund Y. Chao. A biomechanical study of normal functional elbow motion. *Journal of Bone and Joint Surgery (American)*, 63(6):872–877, 1981. URL <https://pubmed.ncbi.nlm.nih.gov/7240327/>.
- [19] David Rempel, Matt J. Camilleri, and David L. Lee. The design of hand gestures for human–computer interaction: Lessons from sign language interpreters. *International Journal of Human-Computer Studies*, 72(10–11):728–735, 2014. doi: 10.1016/j.ijhcs.2014.05.003.
- [20] Jaiyoung Ryu, William P. Cooney, Lawrence J. Askew, Kai-Nan An, and Edmund Y. S. Chao. Functional ranges of motion of the wrist joint. *Journal of Hand Surgery (American)*, 16(3):409–419, 1991. doi: 10.1016/0363-5023(91)90006-w.
- [21] Emma Sala et al. Pinch grip per se is not an occupational risk factor for the musculoskeletal system. *International Journal of Environmental Research and Public Health*, 19(15):8975, 2022. doi: 10.3390/ijerph19158975.
- [22] Marco Santello, Martha Flanders, and John F. Soechting. Postural hand synergies for tool use. *Journal of Neuroscience*, 18(23):10105–10115, 1998. doi: 10.1523/JNEUROSCI.18-23-10105.1998.
- [23] Adwait Sharma et al. SoloFinger: Robust microgestures while grasping everyday objects. In *Proceedings of CHI '21*, 2021. doi: 10.1145/3411764.3445197.
- [24] Hiske van Duinen, Wei Shin Yu, and Simon C. Gandevia. Limited ability to extend the digits of the human hand independently with extensor digitorum. *Journal of Physiology*, 587(20):4799–4810, 2009. doi: 10.1113/jphysiol.2009.177964.
- [25] Nicholas D. Weiss, Leonard Gordon, Todd Bloom, Yuen So, and David M. Rempel. Position of the wrist associated with the lowest carpal-tunnel pressure. *Journal of Bone and Joint Surgery (American)*, 77(11):1695–1699, 1995. doi: 10.2106/00004623-199511000-00008.
- [26] Luke A. Wilhelm, Joel R. Martin, Mark L. Latash, and Vladimir M. Zatsiorsky. Finger enslaving in the dominant and non-dominant hand. *Human Movement Science*, 33:185–193, 2014. doi: 10.1016/j.humov.2013.10.001. Published online 2013.

- [27] Kayo Yin, Terry Regier, and Dan Klein. American Sign Language handshapes reflect pressures for communicative efficiency. In *Proceedings of ACL 2024*, 2024. URL <https://arxiv.org/abs/2406.04024>.
- [28] Wei Shin Yu, Hiske van Duinen, and Simon C. Gandevia. Limits to the control of the human thumb and fingers in flexion and extension. *Journal of Neurophysiology*, 103(1):278–289, 2010. doi: 10.1152/jn.00797.2009.

## A Full canonical ranking

**Table 6:** Canonical ranking with unweighted term costs and the 5–95% rank interval (*model output*). – marks an undefined term.

| #  | Gesture                             | Tier | Cost  | Posture | Coupling | Thumb reach | Endurance | Effort | Everyday pose | Everyday movement | Rank  |
|----|-------------------------------------|------|-------|---------|----------|-------------|-----------|--------|---------------|-------------------|-------|
| 1  | Wrist extend 30°                    | 1    | 0.114 | 0.27    | 0.00     | 0.00        | 0.00      | 0.15   | 0.10          | 0.28              | 1–6   |
| 2  | Forearm pronate 45°                 | 1    | 0.123 | 0.16    | 0.00     | 0.00        | 0.00      | 0.49   | 0.09          | –                 | 1–12  |
| 3  | Wrist flex 30°                      | 1    | 0.124 | 0.28    | 0.00     | 0.00        | 0.00      | 0.15   | 0.16          | 0.28              | 1–5   |
| 4  | Forearm supinate 45°                | 1    | 0.130 | 0.20    | 0.00     | 0.00        | 0.00      | 0.49   | 0.09          | –                 | 1–10  |
| 5  | Thumb out (medium)                  | 1    | 0.155 | 0.23    | 0.10     | 0.00        | 0.00      | 0.10   | 0.12          | 0.53              | 2–9   |
| 6  | Open hand (medium)                  | 1    | 0.162 | 0.12    | 0.40     | 0.00        | 0.00      | 0.41   | 0.09          | 0.12              | 2–17  |
| 7  | Wrist ulnar 20°                     | 1    | 0.165 | 0.28    | 0.00     | 0.00        | 0.00      | 0.20   | 0.09          | 0.59              | 4–11  |
| 8  | Index lift (medium)                 | 1    | 0.177 | 0.09    | 0.12     | 0.00        | 0.00      | 0.14   | 0.35          | 0.55              | 4–11  |
| 9  | Wrist radial 15°                    | 1    | 0.182 | 0.32    | 0.00     | 0.00        | 0.00      | 0.15   | 0.20          | 0.59              | 6–14  |
| 10 | Squeeze fingers together            | 1    | 0.198 | 0.09    | 0.10     | 0.00        | 0.00      | 0.14   | 0.27          | 0.80              | 6–19  |
| 11 | Spread fingers (medium)             | 2    | 0.218 | 0.23    | 0.09     | 0.00        | 0.00      | 0.13   | 0.27          | 0.80              | 9–24  |
| 12 | C handshape                         | 2    | 0.224 | 0.19    | 0.38     | 0.00        | 0.00      | 0.32   | 0.27          | 0.41              | 10–25 |
| 13 | Thumb curl (medium)                 | 1    | 0.232 | 0.15    | 0.11     | 0.00        | 0.00      | 0.13   | 0.79          | 0.45              | 9–22  |
| 14 | Thumb across palm (medium)          | 2    | 0.239 | 0.30    | 0.11     | 0.00        | 0.00      | 0.14   | 0.60          | 0.53              | 12–25 |
| 15 | Knuckle bend (medium)               | 2    | 0.244 | 0.09    | 0.44     | 0.00        | 0.00      | 0.40   | 0.55          | 0.24              | 8–30  |
| 16 | Little lift (medium)                | 2    | 0.252 | 0.09    | 0.45     | 0.00        | 0.00      | 0.12   | 0.45          | 0.67              | 11–35 |
| 17 | Index distal side tap               | 2    | 0.259 | 0.13    | 0.12     | 0.15        | 0.00      | 0.20   | 0.75          | 0.47              | 11–34 |
| 18 | Index curl (medium)                 | 2    | 0.260 | 0.11    | 0.14     | 0.00        | 0.00      | 0.23   | 0.83          | 0.51              | 14–26 |
| 19 | Index distal pad tap                | 2    | 0.262 | 0.13    | 0.11     | 0.07        | 0.00      | 0.22   | 0.77          | 0.53              | 9–33  |
| 20 | Hook (medium)                       | 2    | 0.264 | 0.26    | 0.46     | 0.00        | 0.00      | 0.49   | 0.45          | 0.20              | 12–36 |
| 21 | Index distal side press-hold        | 2    | 0.274 | 0.12    | 0.12     | 0.15        | 0.01      | 0.20   | 0.84          | 0.47              | 13–38 |
| 22 | Index distal pad press-hold         | 2    | 0.276 | 0.13    | 0.12     | 0.07        | 0.01      | 0.22   | 0.85          | 0.53              | 11–37 |
| 23 | O handshape                         | 2    | 0.294 | 0.14    | 0.42     | 0.07        | 0.01      | 0.59   | 0.57          | 0.25              | 17–42 |
| 24 | Index middle side tap               | 2    | 0.298 | 0.13    | 0.11     | 0.30        | 0.00      | 0.26   | 0.81          | 0.46              | 21–41 |
| 25 | Index middle pad tap                | 2    | 0.298 | 0.11    | 0.12     | 0.23        | 0.00      | 0.29   | 0.61          | 0.73              | 15–40 |
| 26 | Index pinch                         | 2    | 0.299 | 0.14    | 0.12     | 0.07        | 0.00      | 0.34   | 0.84          | 0.58              | 9–31  |
| 27 | Index pinch hold                    | 2    | 0.307 | 0.14    | 0.12     | 0.07        | 0.01      | 0.32   | 0.91          | 0.57              | 8–35  |
| 28 | Index middle pad press-hold         | 2    | 0.310 | 0.11    | 0.12     | 0.23        | 0.01      | 0.29   | 0.69          | 0.72              | 18–44 |
| 29 | Index middle side press-hold        | 2    | 0.311 | 0.13    | 0.11     | 0.30        | 0.01      | 0.26   | 0.91          | 0.46              | 23–45 |
| 30 | Index distal pad slide along        | 2    | 0.313 | 0.13    | 0.12     | 0.18        | 0.00      | 0.47   | 0.86          | 0.42              | 19–48 |
| 31 | Index distal side slide along       | 2    | 0.316 | 0.12    | 0.13     | 0.26        | 0.00      | 0.43   | 0.84          | 0.42              | 22–49 |
| 32 | Middle lift (medium)                | 2    | 0.317 | 0.09    | 0.51     | 0.00        | 0.00      | 0.14   | 0.83          | 0.66              | 20–45 |
| 33 | Middle, ring + little curl (medium) | 2    | 0.321 | 0.17    | 0.64     | 0.00        | 0.00      | 0.55   | 0.64          | 0.26              | 20–48 |
| 34 | Index + middle curl (medium)        | 2    | 0.323 | 0.14    | 0.36     | 0.00        | 0.00      | 0.40   | 0.89          | 0.46              | 26–40 |
| 35 | Middle distal pad tap               | 3    | 0.328 | 0.13    | 0.33     | 0.13        | 0.00      | 0.23   | 0.86          | 0.62              | 30–48 |
| 36 | Pinch press                         | 2    | 0.331 | 0.14    | 0.12     | 0.07        | 0.17      | 0.32   | 0.91          | 0.57              | 12–45 |
| 37 | Index distal side slide across      | 3    | 0.331 | 0.13    | 0.14     | 0.28        | 0.00      | 0.45   | 0.87          | 0.46              | 27–51 |
| 38 | Middle middle pad tap               | 3    | 0.338 | 0.15    | 0.31     | 0.30        | 0.00      | 0.31   | 0.79          | 0.50              | 21–46 |
| 39 | Middle distal pad press-hold        | 3    | 0.341 | 0.12    | 0.33     | 0.13        | 0.01      | 0.23   | 0.94          | 0.62              | 33–52 |
| 40 | Index middle side slide across      | 3    | 0.344 | 0.13    | 0.12     | 0.45        | 0.00      | 0.34   | 0.92          | 0.46              | 30–59 |
| 41 | Middle pinch                        | 3    | 0.345 | 0.13    | 0.37     | 0.13        | 0.00      | 0.31   | 0.88          | 0.61              | 36–49 |
| 42 | Close hand (medium)                 | 3    | 0.348 | 0.35    | 0.47     | 0.00        | 0.00      | 0.69   | 0.74          | 0.19              | 25–55 |
| 43 | Middle curl (medium)                | 3    | 0.350 | 0.11    | 0.57     | 0.00        | 0.00      | 0.23   | 0.89          | 0.65              | 28–53 |
| 44 | Little curl (medium)                | 3    | 0.350 | 0.12    | 0.51     | 0.00        | 0.00      | 0.23   | 0.91          | 0.69              | 30–54 |
| 45 | Middle middle pad press-hold        | 3    | 0.353 | 0.15    | 0.31     | 0.30        | 0.01      | 0.31   | 0.89          | 0.50              | 23–50 |
| 46 | Middle pinch hold                   | 3    | 0.355 | 0.13    | 0.36     | 0.13        | 0.01      | 0.30   | 0.95          | 0.61              | 39–52 |
| 47 | Ring + little curl (medium)         | 3    | 0.365 | 0.15    | 0.68     | 0.00        | 0.00      | 0.41   | 0.85          | 0.47              | 32–56 |
| 48 | Ring lift (medium)                  | 3    | 0.366 | 0.09    | 0.73     | 0.00        | 0.00      | 0.11   | 0.89          | 0.74              | 25–60 |
| 49 | 5 handshape                         | 3    | 0.378 | 0.62    | 0.40     | 0.00        | 0.00      | 0.69   | 0.70          | 0.23              | 32–67 |
| 50 | Index middle pad slide along        | 3    | 0.378 | 0.11    | 0.13     | 0.37        | 0.00      | 0.64   | 0.86          | 0.53              | 33–62 |
| 51 | Middle distal pad slide along       | 3    | 0.391 | 0.13    | 0.44     | 0.20        | 0.00      | 0.44   | 0.95          | 0.58              | 47–63 |
| 52 | Ring distal pad tap                 | 3    | 0.392 | 0.18    | 0.47     | 0.23        | 0.00      | 0.37   | 0.85          | 0.64              | 48–65 |
| 53 | Index middle side slide along       | 3    | 0.401 | 0.13    | 0.13     | 0.45        | 0.00      | 0.72   | 0.94          | 0.45              | 37–68 |
| 54 | Ring curl (medium)                  | 3    | 0.406 | 0.12    | 0.84     | 0.00        | 0.00      | 0.24   | 0.89          | 0.75              | 37–68 |
| 55 | Ring distal pad press-hold          | 3    | 0.407 | 0.17    | 0.47     | 0.23        | 0.01      | 0.37   | 0.95          | 0.64              | 51–69 |
| 56 | A handshape                         | 3    | 0.407 | 0.35    | 0.55     | 0.00        | 0.00      | 0.85   | 0.94          | 0.15              | 36–68 |
| 57 | Little distal pad tap               | 3    | 0.409 | 0.25    | 0.25     | 0.42        | 0.00      | 0.43   | 0.91          | 0.60              | 49–69 |
| 58 | Ring pinch                          | 3    | 0.409 | 0.17    | 0.54     | 0.23        | 0.00      | 0.37   | 0.91          | 0.65              | 50–65 |

| #  | Gesture                       | Tier | Cost  | Posture | Coupling | Thumb reach | Endurance | Effort | Everyday pose | Everyday movement | Rank  |
|----|-------------------------------|------|-------|---------|----------|-------------|-----------|--------|---------------|-------------------|-------|
| 59 | Little pinch                  | 3    | 0.413 | 0.22    | 0.28     | 0.42        | 0.00      | 0.38   | 0.96          | 0.62              | 50-68 |
| 60 | Ring pinch hold               | 3    | 0.414 | 0.16    | 0.54     | 0.23        | 0.01      | 0.37   | 0.94          | 0.65              | 52-67 |
| 61 | Little pinch hold             | 3    | 0.415 | 0.20    | 0.28     | 0.42        | 0.01      | 0.38   | 0.97          | 0.63              | 50-69 |
| 62 | Little distal pad press-hold  | 3    | 0.416 | 0.23    | 0.25     | 0.42        | 0.01      | 0.43   | 0.97          | 0.60              | 52-72 |
| 63 | Ring middle pad tap           | 3    | 0.424 | 0.17    | 0.40     | 0.47        | 0.00      | 0.42   | 0.88          | 0.62              | 57-74 |
| 64 | B handshape                   | 3    | 0.424 | 0.62    | 0.40     | 0.00        | 0.00      | 0.66   | 0.94          | 0.36              | 49-71 |
| 65 | Middle middle pad slide along | 3    | 0.432 | 0.15    | 0.36     | 0.34        | 0.00      | 0.67   | 0.93          | 0.57              | 51-68 |
| 66 | Ring middle pad press-hold    | 4    | 0.436 | 0.16    | 0.40     | 0.47        | 0.01      | 0.42   | 0.96          | 0.62              | 61-76 |
| 67 | S handshape                   | 3    | 0.441 | 0.55    | 0.54     | 0.00        | 0.00      | 0.86   | 0.96          | 0.19              | 46-72 |
| 68 | Little middle pad tap         | 4    | 0.449 | 0.21    | 0.30     | 0.60        | 0.00      | 0.49   | 0.94          | 0.60              | 58-78 |
| 69 | Little middle pad press-hold  | 4    | 0.454 | 0.19    | 0.30     | 0.60        | 0.01      | 0.49   | 0.98          | 0.60              | 59-79 |
| 70 | L handshape                   | 3    | 0.459 | 0.42    | 0.63     | 0.00        | 0.00      | 0.82   | 0.96          | 0.38              | 56-73 |
| 71 | Fist squeeze                  | 3    | 0.471 | 0.55    | 0.54     | 0.00        | 0.17      | 0.88   | 0.96          | 0.19              | 54-79 |
| 72 | Little distal pad slide along | 4    | 0.479 | 0.23    | 0.33     | 0.61        | 0.00      | 0.62   | 0.97          | 0.60              | 66-81 |
| 73 | Y handshape                   | 4    | 0.488 | 0.47    | 0.57     | 0.00        | 0.00      | 0.81   | 0.98          | 0.58              | 64-76 |
| 74 | I handshape                   | 4    | 0.488 | 0.58    | 0.62     | 0.00        | 0.00      | 0.82   | 0.97          | 0.42              | 61-77 |
| 75 | Ring distal pad slide along   | 4    | 0.489 | 0.17    | 0.63     | 0.44        | 0.00      | 0.58   | 0.95          | 0.64              | 68-81 |
| 76 | Ring middle pad slide along   | 4    | 0.492 | 0.16    | 0.50     | 0.52        | 0.00      | 0.70   | 0.96          | 0.61              | 68-81 |
| 77 | I handshape                   | 4    | 0.503 | 0.58    | 0.57     | 0.00        | 0.00      | 0.82   | 0.98          | 0.57              | 67-79 |
| 78 | 3 handshape                   | 4    | 0.504 | 0.55    | 0.63     | 0.00        | 0.00      | 0.79   | 0.98          | 0.58              | 67-79 |
| 79 | W handshape                   | 4    | 0.505 | 0.62    | 0.50     | 0.00        | 0.00      | 0.72   | 0.99          | 0.70              | 71-81 |
| 80 | V handshape                   | 4    | 0.519 | 0.62    | 0.63     | 0.00        | 0.00      | 0.80   | 0.98          | 0.60              | 72-80 |
| 81 | Horns                         | 4    | 0.527 | 0.62    | 0.65     | 0.00        | 0.00      | 0.78   | 0.98          | 0.66              | 74-81 |

## B Unreachable contacts

The rig’s thumb misses these targets by more than 5 mm: Index proximal pad tap (13 mm), Index proximal pad press-hold (13 mm), Index proximal pad slide along (20 mm), Middle proximal pad tap (18 mm), Middle proximal pad press-hold (18 mm), Middle proximal pad slide along (23 mm), Ring proximal pad tap (23 mm), Ring proximal pad press-hold (23 mm), Ring proximal pad slide along (28 mm), Little middle pad slide along (8 mm), Little proximal pad tap (18 mm), Little proximal pad press-hold (18 mm), Little proximal pad slide along (22 mm), Index proximal side tap (16 mm), Index proximal side press-hold (16 mm), Index proximal side slide along (24 mm), Index proximal side slide across (16 mm) . DigitSpace rates proximal segments the least comfortable (2.1–2.7 of 5) [9], so their exclusion is unlikely to remove natural gestures.

## C Assumptions

- HOME: Ingram et al.’s mean finger posture; thumb resting beside the index; neutral wrist and forearm; spread 0.25.
- Amplitudes: 30/60/100% of the way to the end shape; wrist flexion and extension 15/30/50°, radial 8/15/25°, ulnar 10/20/35°, forearm 20/45/70°.
- Contact solver: everyday Mahalanobis prior at  $10^{-4}$  per unit (16 units per mm of miss); touched-finger curl  $1.6 \times 10^{-5}$  per degree<sup>2</sup>; reach threshold 5 mm.
- Posture: shape weight 0.65, wrist 0.85; 70% peak and 30% mean; Rempel’s deviation ratings assumed to apply at 20° radial and 30° ulnar; LUBA classes scaled to 0.6; the thumb ramps to 3.2 at 50° across the palm and to 2.0 at 35° out.
- Coupling: 0.55 leak and 0.45 discordance; leak scale 0.10; extension factor 1.5 for the middle, ring and little fingers.
- Thumb reach: tip contacts use the distal-segment rating; side penalty 0.08.
- Endurance: 60 uses; 0.02, 0.04 and 0.25 of maximum effort.
- Effort: weights 1, 2 and 3; scale 4 range-motions.

- Everyday terms: ridge 1% of the largest velocity eigenvalue; wrist artefact limits  $-80$  to  $80^\circ$  flexion and  $-45$  to  $30^\circ$  deviation.
- Combinations: 25% core overlap for sequences; chord clutch held for the full performance of B.
- Coupling arc factor 1 (varied 0–2).
- Uncertainty draws: Dirichlet concentration 5 per term; assumption grids as in Table 4; 200 subject bootstraps; 20 IK replicates.

## D Reproducibility

- `.venv-analysis/bin/python-Ianalysis/everyday_hand.py` (and `--no-limits`) computes the everyday statistics from `KIN_MUS_UJI.mat`.
- `node scripts/natural/run.mjs` samples, solves and scores every gesture and combination and runs every experiment (`docs/paper/data/results.json`).
- `node scripts/natural/paper-tables.mjs` and `analysis/figures.py` render the tables and figures.
- `node scripts/natural/check.mjs` verifies the data, model invariants and outputs.

The interactive demonstration at <https://ethanlambeth.com/natural-hand/> plays every gesture and combination and shows its cost breakdown.